

Analyzing Economic Factors that Affect Tech Companies' Revenue Growth

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Group Evaluation:

Everyone in the group has worked very hard on this project. Each one of us did our part equally, so we each contributed about 25% of the whole project. I wrote the codes for the models on R Studio, made the regression summaries, and wrote the methodology. Alexa wrote the abstract and introduction for this report. Sasha collected the data and analyzed the results, while Hamza helped write the conclusion. Everyone of us worked on the slides although Alexa laid out the groundwork in terms of format and outlines. This does not necessarily mean any one of us just wrote that section individually as we all discussed together while writing those sections, so for example, when I wrote the methodology section, the rest of the group also helped me come up with the appropriate formula. Overall, no one is really overdoing any work, considering both the slides and this final report.

Abstract

The technology sector, a continuously expanding industry, spearheads innovation, job creation, and investment. Our aim was to explore how this sector interacts with the broader economy. We employed a linear regression analysis in R to examine the revenue growth of the top 10 technology companies listed in the S&P 500 over the past decade, alongside various economic indicators. Our analysis revealed that these top companies operated independently of economic factors such as CPI inflation rate, GDP, and private gross investment. None of the probability values were significant, suggesting a lack of correlation between these variables and the performance of the technology companies in our model.

Introduction

Our study looks into the intricate relationship between the technology sector and broader economic conditions. We concentrated on the top 10 technology firms listed in the S&P 500, including industry giants like Microsoft, Apple, Oracle, and others. Our primary objective was to discern whether the revenue growth of these companies was influenced by concurrent economic factors. To accomplish this, we gathered data from the Federal Reserve Economic Data (FRED), capturing key economic indicators such as the Nominal GDP growth rate, Interest rate, Unemployment rate, and more. This comprehensive dataset enabled us to explore how fluctuations in these economic metrics might correlate with the financial performance of the selected tech companies.

Using a Linear Regression model, we analyzed the collected data to identify any potential connections between the revenue growth patterns of the tech firms and economic patterns. Our aim was to shed light on whether the technology sector operates independently of broader economic shifts or whether it is significantly influenced by them. Through our research, we aimed to provide valuable insights into the resilience and adaptability of the technology industry amidst changing economic landscapes. By understanding these dynamics, stakeholders can make more informed decisions regarding investments and strategic planning within the tech sector. Our findings happened to be that the technology sector does in fact stand unaffected by economic structures. When the regressions were run through R, we saw little significance in our coefficients and low probability values. The rest of this paper is split into the following sections: Data collection, Methodology, and Results/Conclusion.

Data Collection

The Federal Reserve Economic Data (FRED) was the source for the data collected for the economic factors: Consumer Price Index (CPI), nominal GDP growth rate, interest rate, unemployment rate and gross private domestic investment (domestic business, Consumer confidence index. The revenue growth of the selected companies were sourced from Stock Analysis.

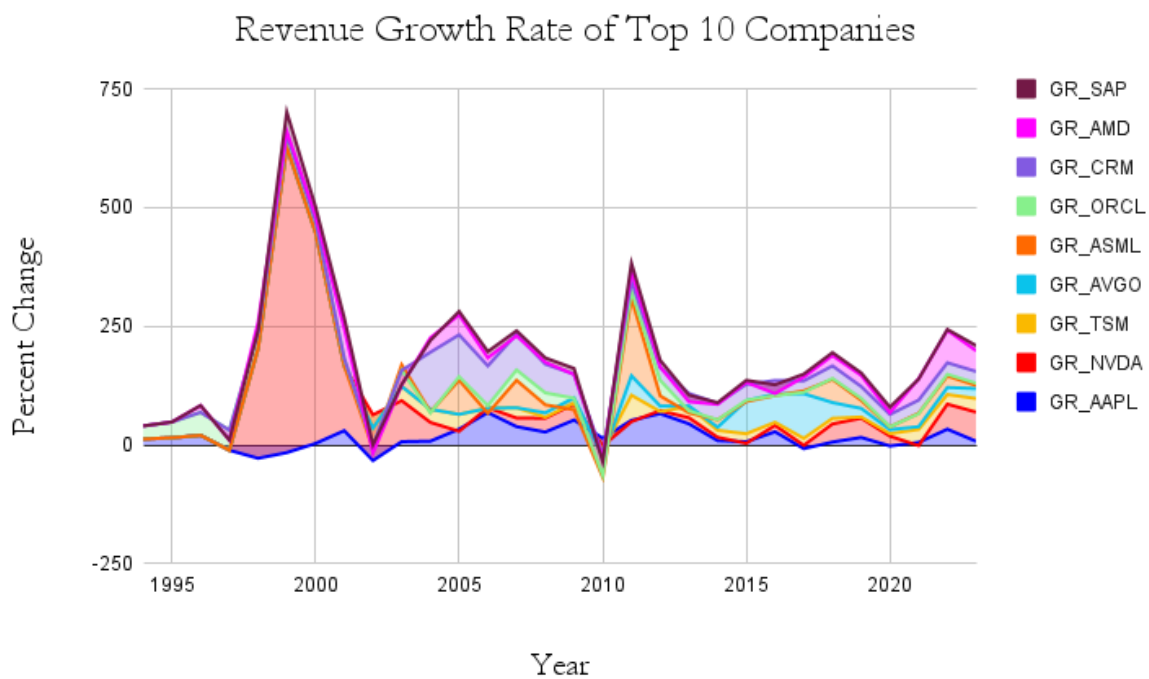


Figure 1: Stacked area chart showing the revenue growth of each company

Methodology

The linear regression was chosen due to its ability to quantify and model the linear dependencies between multiple variables, making it well-suited for exploring the complex interplay between company performance and economic trends. The following is the empirical formula:

$Revenue\ Growth\ Rate = \beta_0 + \beta_1 CPI + \beta_2 NGDP + \beta_3 IR + \beta_4 UNRATE + \beta_5 GPDI + \varepsilon$, where CPI stands for the Consumer Price Index; the NGDP stands for the nominal GDP growth rate; the IR stands for the interest rates or the federal funds rate; UNRATE is the unemployment rate; and the GPDI is the gross domestic investment rate. Each of the economic indicators are multiplied by their respective coefficients in the model.

Results

The summary snapshots below showcase the outcomes of interpreted linear regression models. Provided are conclusive insights underneath each regression summary for each of the companies. Located at the end of the data section is a chart indicating the significance or insignificance of coefficients and probability values.

```
> summary(reg_MSFT)

Call:
lm(formula = GR_MSFT ~ CPI + NGDP + IR + UNRATE + GPDI, data = ts_data)

Residuals:
    Min       1Q   Median       3Q      Max
-13.6496  -4.7494   0.4907   3.2953  19.1008

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  -5.4334    13.2804  -0.409  0.68593
CPI             0.6929     2.2021   0.315  0.75565
NGDP          -0.2106     2.7255  -0.077  0.93903
IR              3.5696     0.9656   3.697  0.00107 **
UNRATE         1.5332     1.2731   1.204  0.23976
GPDI           0.4779     0.2287   2.089  0.04702 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 8.629 on 25 degrees of freedom
Multiple R-squared:  0.5051,    Adjusted R-squared:  0.4061
F-statistic: 5.103 on 5 and 25 DF,  p-value: 0.002325
```

Figure 2.1: Regression summary for Microsoft (MSFT)

```
> summary(reg_AAPL)

Call:
lm(formula = GR_AAPL ~ CPI + NGDP + IR + UNRATE + GPDI, data = ts_data)

Residuals:
    Min       1Q   Median       3Q      Max
-35.230 -14.443   0.439  11.410  52.384

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  -3.81535    33.58836  -0.114  0.9105
CPI           11.82575     5.56954   2.123  0.0438 *
NGDP          -9.76626     6.89322  -1.417  0.1689
IR              0.03354     2.44212   0.014  0.9892
UNRATE         5.59696     3.21993   1.738  0.0945 .
GPDI           0.67960     0.57852   1.175  0.2512
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 21.82 on 25 degrees of freedom
Multiple R-squared:  0.3501,    Adjusted R-squared:  0.2201
F-statistic: 2.693 on 5 and 25 DF,  p-value: 0.04432
```

Figure 2.2: Regression summary for Apple (AAPL)

```
> summary(reg_NVDA)

Call:
lm(formula = GR_NVDA ~ CPI + NGDP + IR + UNRATE + GPDI, data = ts_data)

Residuals:
    Min       1Q   Median       3Q      Max
-132.78  -54.18  -11.45   38.91  360.19

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  -190.106    184.037  -1.033   0.3134
CPI           -74.591     31.488  -2.369   0.0275 *
NGDP           71.414     39.514   1.807   0.0851 .
IR             27.676     16.422   1.685   0.1067
UNRATE        7.543      18.376   0.410   0.6856
GPDI          3.404       3.274   1.040   0.3103
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 117.7 on 21 degrees of freedom
(4 observations deleted due to missingness)
Multiple R-squared:  0.4868,    Adjusted R-squared:  0.3646
F-statistic: 3.984 on 5 and 21 DF,  p-value: 0.01067
```

Figure 2.3: Regression summary for Nvidia (NVDA)

```
> summary(reg_TSM)

Call:
lm(formula = GR_TSM ~ CPI + NGDP + IR + UNRATE + GPDI, data = ts_data)

Residuals:
    Min       1Q   Median       3Q      Max
-30.381  -8.249  -1.234   11.242   21.508

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  -6.3033    26.9764  -0.234   0.818
CPI           -0.2106     4.9654  -0.042   0.967
NGDP           0.5235     5.8496   0.090   0.930
IR            -1.8434     2.7601  -0.668   0.513
UNRATE        2.8088     2.5997   1.080   0.295
GPDI          1.1074     0.4803   2.306   0.034 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 15.94 on 17 degrees of freedom
(8 observations deleted due to missingness)
Multiple R-squared:  0.3978,    Adjusted R-squared:  0.2206
F-statistic: 2.246 on 5 and 17 DF,  p-value: 0.09676
```

Figure 2.4: Regression summary for Taiwan Semiconductor (TSM)

```
> summary(reg_AVGO)

Call:
lm(formula = GR_AVGO ~ CPI + NGDP + IR + UNRATE + GPDI, data = ts_data)

Residuals:
    Min       1Q   Median       3Q      Max
-32.794 -14.898  -4.878  12.293  43.174

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 136.3473   49.0367   2.781  0.0179 *
CPI           5.3746   10.0130   0.537  0.6021
NGDP        -14.1777   11.9169  -1.190  0.2592
IR           -8.1740    5.3306  -1.533  0.1534
UNRATE      -9.9683    4.3315  -2.301  0.0419 *
GPDI          0.7504    0.8325   0.901  0.3867
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 24.77 on 11 degrees of freedom
(14 observations deleted due to missingness)
Multiple R-squared:  0.4507,    Adjusted R-squared:  0.201
F-statistic: 1.805 on 5 and 11 DF,  p-value: 0.1924
```

Figure 2.5: Regression summary for Broadcom (AVGO)

```
> summary(reg_ASML)

Call:
lm(formula = GR_ASML ~ CPI + NGDP + IR + UNRATE + GPDI, data = ts_data)

Residuals:
    Min       1Q   Median       3Q      Max
-72.263 -20.226   0.074  17.671  82.968

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -15.333    65.495  -0.234  0.81769
CPI           5.230    12.055   0.434  0.66986
NGDP        -13.220    14.202  -0.931  0.36496
IR           10.468     6.701   1.562  0.13666
UNRATE        8.161     6.312   1.293  0.21332
GPDI          3.447     1.166   2.956  0.00884 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 38.69 on 17 degrees of freedom
(8 observations deleted due to missingness)
Multiple R-squared:  0.3805,    Adjusted R-squared:  0.1984
F-statistic: 2.089 on 5 and 17 DF,  p-value: 0.1169
```

Figure 2.6: Regression summary for ASML Holding (ASML)


```
> summary(reg_ORCL)

Call:
lm(formula = GR_ORCL ~ CPI + NGDP + IR + UNRATE + GPDI, data = ts_data)

Residuals:
    Min       1Q   Median       3Q      Max
-12.2742  -4.9714   0.1618   4.8867  10.9857

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -12.3009    10.5272  -1.168  0.253632
CPI           7.4200     1.7456   4.251  0.000260 ***
NGDP          -8.2993     2.1605  -3.841  0.000744 ***
IR            7.9104     0.7654  10.335 1.64e-10 ***
UNRATE        4.2022     1.0092   4.164  0.000325 ***
GPDI          0.4342     0.1813   2.395  0.024441 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6.84 on 25 degrees of freedom
Multiple R-squared:  0.8261,    Adjusted R-squared:  0.7913
F-statistic: 23.75 on 5 and 25 DF,  p-value: 9.42e-09
```

Figure 2.7: Regression summary for Oracle (ORCL)

```
> summary(reg_CRM)

Call:
lm(formula = GR_CRM ~ CPI + NGDP + IR + UNRATE + GPDI, data = ts_data)

Residuals:
    Min       1Q   Median       3Q      Max
-39.957 -12.607  -4.635   1.312  78.216

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) -36.1986    50.2866  -0.720   0.483
CPI          -16.5798     9.8584  -1.682   0.113
NGDP          19.1918    11.8683   1.617   0.127
IR             6.3065     4.9519   1.274   0.222
UNRATE        4.7649     4.6588   1.023   0.323
GPDI           0.1602     0.8928   0.179   0.860

Residual standard error: 28.37 on 15 degrees of freedom
(10 observations deleted due to missingness)
Multiple R-squared:  0.2491,    Adjusted R-squared:  -0.00119
F-statistic: 0.9952 on 5 and 15 DF,  p-value: 0.4534
```

Figure 2.8: Regression summary for Salesforce (CRM)

```
> summary(reg_AMD)
```

Call:
lm(formula = GR_AMD ~ CPI + NGDP + IR + UNRATE + GPDI, data = ts_data)

Residuals:

	Min	1Q	Median	3Q	Max
	-40.456	-13.610	1.232	14.406	42.630

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-40.0049	35.8148	-1.117	0.276
CPI	-1.2417	5.8638	-0.212	0.834
NGDP	11.1347	7.2682	1.532	0.139
IR	-3.1855	2.6636	-1.196	0.244
UNRATE	1.4122	3.5073	0.403	0.691
GPDI	0.4252	0.6286	0.676	0.506

Residual standard error: 22.94 on 23 degrees of freedom
(2 observations deleted due to missingness)
Multiple R-squared: 0.2937, Adjusted R-squared: 0.1402
F-statistic: 1.913 on 5 and 23 DF, p-value: 0.1311

Figure 2.9: Regression summary for Advanced Micro Devices (AMD)

```
> summary(reg_SAP)
```

Call:
lm(formula = GR_SAP ~ CPI + NGDP + IR + UNRATE + GPDI, data = ts_data)

Residuals:

	Min	1Q	Median	3Q	Max
	-33.074	-4.382	-1.336	5.631	25.366

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	6.35266	17.65065	0.360	0.723
CPI	-0.26805	3.01995	-0.089	0.930
NGDP	-0.54863	3.78972	-0.145	0.886
IR	1.51926	1.57500	0.965	0.346
UNRATE	0.05919	1.76243	0.034	0.974
GPDI	0.39501	0.31398	1.258	0.222

Residual standard error: 11.29 on 21 degrees of freedom
(4 observations deleted due to missingness)
Multiple R-squared: 0.1324, Adjusted R-squared: -0.07423
F-statistic: 0.6407 on 5 and 21 DF, p-value: 0.6713

Figure 2.10: Regression summary for SAP SE (SAP)

The table below summarizes the findings from the regression analyses above. A check in any box means that variable is statistically significant in influencing the revenue growth rate for a company:

	Statistically Significant or Not				
Company Name	Consumer Price Index	Nominal GDP Growth Rate	Interest Rate	Unemployment Rate	Gross Private Domestic Investment
MSFT	×	×	✓	×	✓
AAPL	✓	×	×	×	×
NVDA	✓	×	×	×	×
TSM	×	×	×	×	✓
AVGO	×	×	×	✓	×
ASML	×	×	×	×	✓
ORCL	✓	✓	✓	✓	✓
CRM	×	×	×	×	×
AMD	×	×	×	×	×
SAP	×	×	×	×	×

The gross private domestic investment is statistically significant in four companies. There are two companies that are affected by the unemployment rate, two by the interest rate, one by the nominal GDP growth rate, and three by the Consumer Price Index. In other words, the gross private domestic investment is the most important feature relative to other variables, which is followed by the CPI. The least important feature is the nominal GDP rate.

Oracle was the only company that had statistical significance in all five macroeconomic factors in the regression. The interest rate and the private investment are statistically significant for Microsoft. The CPI was the only factor that was statistically significant for Apple and Nvidia.

Next, the gross private domestic investment was the only factor that was statistically significant for TSM and ASML. AVGO only had one macroeconomic factor that was statistically significant, the unemployment. Lastly, the macroeconomic factors in the regression have no statistically significance for CRM, AMD, and SAP.

We have included charts displaying the correlation between each of the companies we studied and the economic indicators. We have also included the R-squared metric for the relationships for the companies which have data for all the years we observed. A higher R-squared generally means that more of the data can be explained by the model we are using. To imply statistical significance, we are looking for R-squared measures on the higher end of the range 0 to 1. The majority of R-squared figures we attained were on the lower end of the spectrum, which implies lower correlation. The only exception to this is for Oracle as its adjusted R-squared is close to 0.8, which can be explained by the fact that all of the independent variables are statistically significant for this company. We generally saw slightly higher measures for gross private domestic investment among the companies we studied, which implies that changes in gross private domestic investment correlate more with revenue growth than the other economic indicators. However, the R-squared measure generally was not useful in our analysis since the majority of the variables were not statistically significant.

Conclusion

According to our observations and analysis, the economic indicators we studied had influence over the revenue growth of the largest technology companies in the United States. The economic metric which had the largest influence over the growth numbers was gross domestic private investments. This is a reasonable conclusion to make as large technology companies rely

heavily on private funding, so they have greater potential to flourish during times when investment is higher. The nominal GDP growth rate turned out to be the least statistically significant measure. Although the technology sector is a key aspect of the macroeconomy, this industry has more volatility and higher growth rates than other sectors of the economy, which is one of the reasons why it may have a lower correlation with nominal GDP growth rate.

We can use our findings to infer the impact that these economic indicators would have on the largest firms in other sectors of the economy. It is reasonable to predict that these firms would also be impacted more heavily by gross private domestic investments, as that is a key source of growth for large corporations. We predict that interest rates will have a greater impact on other sectors, especially those which rely on lending and debt financing.

Due to the limited amount of the data that we were working with, our findings were informative but we could get more conclusive results if we include more companies in our dataset. This is particularly true for companies that are toward the bottom of the list since they are smaller in size, which means we can assume their revenue growth would be much more affected by these economic factors.

References

A list of stocks in the technology sector. Stock Analysis. (n.d.). <https://stockanalysis.com/stocks/sector/technology/>

Consumer price index for all urban consumers: All items in U.S. city average. FRED. (2024, April 10). <https://fred.stlouisfed.org/series/CPIAUCSL>

Nominal gross domestic product for United States. FRED. (2024b, February 12). <https://fred.stlouisfed.org/series/NGDPSAXDCUSQ>

Federal funds effective rate. FRED. (2024b, May 1). <https://fred.stlouisfed.org/series/FEDFUNDS>

Unemployment rate. FRED. (2024d, May 3). <https://fred.stlouisfed.org/series/UNRATE>

Gross Private Domestic Investment. FRED. (2024c, April 25). <https://fred.stlouisfed.org/series/GPDI>